

Degradable Polymer with Protein Resistance in a Marine Environment

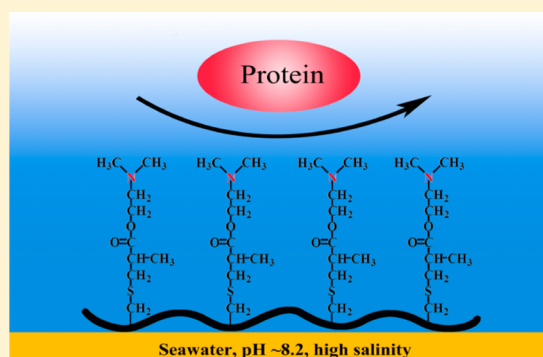
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S Supporting Information

ABSTRACT: Protein resistance is the central issue in marine antibiofouling. We have prepared poly(ϵ -caprolactone) (PCL)-based polyurethane with 2-(dimethylamino) ethyl methacrylate (DEM) as pendant groups by a combination of the thiol–ene click reaction and the condensation reaction. By the use of quartz crystal microbalance with dissipation (QCM-D) and surface plasmon resonance (SPR), we have investigated the adsorption of fibrinogen, bovine serum albumin (BSA), and lysozyme on the polymer surface. The polymer exhibits protein resistance in seawater but not in fresh water because DEM pendant groups carry net neutral charges in the former. The evaluation of antibacterial adhesion of the polymer by using *Micrococcus luteus* demonstrates that the polymer can effectively inhibit the settlement of marine bacteria. Our studies also show that the polymer is degradable in marine environments.



1. INTRODUCTION

Nonspecific protein adsorption onto a surface plays a critical role in a variety of biologically related processes and applications. Marine biofouling usually starts from the formation of a biofilm due to the adsorption of microorganisms consisting of proteins and glycoproteins on underwater surfaces.^{1,2} The adsorption of proteins on membranes and within pores used in seawater desalination and biochemical product separations would degrade their performance.^{3,4} The unwanted protein adsorption can also sharply reduce the sensitivity of biosensors.⁵ Thus, protein-resistant materials such as polysaccharides,⁶ poly(vinyl alcohol),⁷ poly(*N*-vinylpyrrolidone),⁸ poly(2-oxazoline),^{9,10} poly(ethylene glycol) (PEG),^{11,12} and zwitterionic polymers^{13,14} have been developed in past years. Generally, they are hydrophilic with net neutral charges, and their protein resistance is attributed to the hydration layer which can effectively reduce the hydrophobic interactions between the proteins and polymer surface.^{15,16} If the hydrophilic polymers have net charges, then they may adsorb proteins due to electrostatic attractions.^{17–19} Note that the hydrophilic protein-resistant polymers usually have a low mechanical strength because they are swollen in aqueous solution.^{20,21} Moreover, they are generally not biodegradable, which has a negative effect on marine environments.

In the present study, we have prepared polyurethane with poly(ϵ -caprolactone) (PCL) segments in the main chain and 2-(dimethylamino) ethyl methacrylate (DEM) as pendant groups. The combination of PU and DEM groups is expected to increase the mechanical strength and coating ability.²² The

DEM groups are hydrophilic but positively charged with a pK_a of ~ 7.4 in fresh water.²³ Yet, as we report, the polymer does not carry net charges in marine environments so that it exhibits protein resistance. Furthermore, the polymer with PCL segments is biodegradable. The polymer is expected to develop into environmentally friendly marine antibiofouling materials.

2. EXPERIMENTAL SECTION

2.1. Materials. Poly(ϵ -caprolactone) diol (PCL, $M_w = 2000$ g/mol) was from Perstorp and was dried under reduced pressure for 2 h prior to use. 2-(Dimethylamino) ethyl methacrylate (DEM), 3-mercaptopropanediol (TG), isophorone diisocyanate (IPDI), and 2,2-dimethoxy-2-phenylacetophenone (DMPA) were from Aladdin. 1,4-Butane diol (1,4-BD), dibutyltin dilaurate (DBTDL), and lipase from *Pseudomonas cepacia* (PS) were from Aldrich. Fibrinogen (fraction I from human plasma), lysozyme via chicken egg white, and BSA (fraction V from the crystalline form) were purchased from Merck Chemicals. DEM and 1,4-BD were purified under reduced pressure before use. Tetrahydrofuran (THF) was refluxed over CaH_2 and distilled prior to use. Artificial seawater (ASW) was prepared following ASTM D1141-98 (2003).²⁴ The pH of ASW or fresh water was mediated by adding 1.0 M HCl or a NaOH aqueous solution. Each protein solution (1.0 mg/mL) was prepared by dissolving the protein in ASW or fresh water. Other reagents were used as received. The synthesis of biodegradable polyurethane with DEM pendant groups is illustrated in Scheme 1.

2.2. Synthesis of DEM(OH)₂. DEM(OH)₂ was prepared via the thiol–ene click reaction.²⁵ Typically, 2.16 g of TG (20 mmol), 1.57 g

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Scheme 1. Synthesis of Biodegradable Polyurethane with Protein Resistance

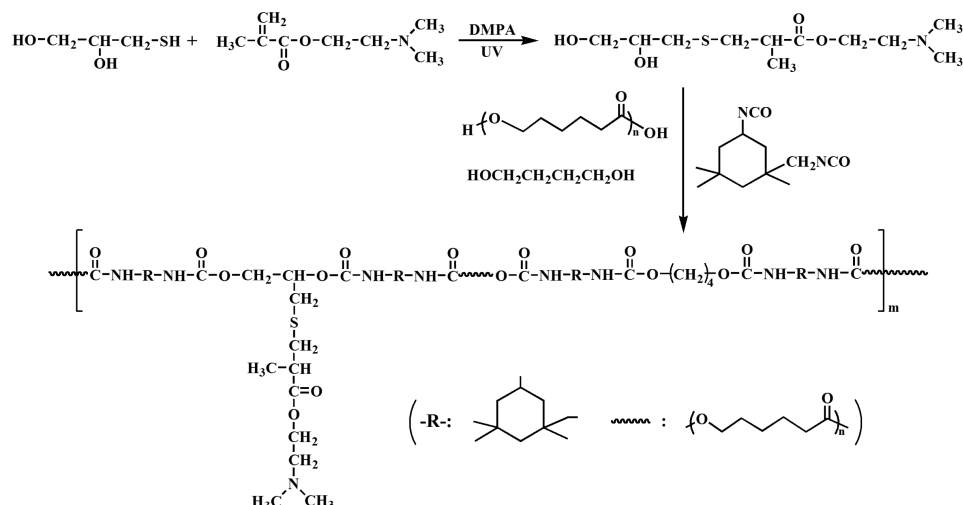


Table 1. Characterization Data of PU-Dx

sample	IPDI/PCL/DEM(OH) ₂ / 1,4-BD ^a	DEM content ^b	<i>M_n</i> ^c	PDI ^c	Δ <i>H_m</i> (J·g ⁻¹) ^d	<i>C_r</i> (%) ^d
PU-D0	37/50/0/13	0	49 000	1.3	21.5	31.6
PU-D7	35/50/5/10	7.4	31 000	1.5	19.7	29.0
PU-D16	30/50/15/5	16.2	32 000	1.6	15.0	22.0
PU-D25	26/50/24/0	24.8	34 000	1.4	4.2	6.2

^aFeed weight ratio. ^bDetermined by elemental analysis. ^cDetermined by GPC. ^dCrystallinity (*C_r*) is determined by DSC.

of DEM (10 mmol), and 0.0383 g of DMPA (1 wt % based on the total mass of monomers) were dissolved in 15.0 mL of THF. The solution was degassed with nitrogen for 20 min. The sample was incubated by irradiation with a UV lamp (emitting nominally at 350 nm, light intensity 100%, 100 mW/cm²) for 2 h. The product was precipitated into hexane twice, filtered, and dried under vacuum at 40 °C for 24 h. ¹H NMR (600 MHz, CDCl₃, ppm): 3.80 (HOCH₂CH₂), 3.57 (CHOHCH₂S), 2.72 and 2.47 (CH₂OHCH₂S), 2.80 (SCH₂CH(CH₃)), 2.66 (SCH₂CH(CH₃)C), 1.25 (CH₂C(CH₃)), 4.16 (COOCH₂CH₂), 2.63 (COOCH₂CH₂), 2.29 (N(CH₃)₂). FTIR (cm⁻¹): 3370 (OH), 2940 (CH₂), 2830 and 2780 (CH₃), 1730 (C=O). HRMS *m/z*: 266.1421 (calculated for C₁₁H₂₄NO₄S: 266.1424).

2.3. Synthesis of Polyurethane with DEM Pendant Groups.

Polyurethane with DEM pendant groups was synthesized via a condensation reaction in THF under a nitrogen atmosphere. The reaction of IPDI and PCL was conducted at 70 °C for 1 h, yielding a prepolymer. Subsequently, DEM(OH)₂ was introduced, and the reaction was conducted at 70 °C for another 1 h. Finally, 1,4-BD and DBTDL were added as the chain extender and catalyst, respectively, and the mixture was allowed to react at 80 °C for 3 h. The resultant polyurethane was precipitated into excess water and dried under vacuum for 24 h. ¹H NMR (600 MHz, CDCl₃, ppm): 4.18 (OCH₂CH₂N(CH₃)₂), 2.65 (OCH₂CH₂N(CH₃)₂), 2.26 (OCH₂CH₂N(CH₃)₂), 1.38 (NCH₂C(CH₃)CH₃), 1.06 (NCH₂C(CH₃)CH₂C(CH₃)₂), 4.05 (COCH₂CH₂CH₂CH₂O), 2.30 (COCH₂CH₂CH₂CH₂O), 1.65 (COCH₂CH₂CH₂CH₂O), 1.38 (COCH₂CH₂CH₂CH₂O), 3.68 (OCH₂CH₂CH₂CH₂O), 1.63 (OCH₂CH₂CH₂CH₂O). IR (cm⁻¹): 3360 (NH), 2950 (CH₃), 2860 (CH₂), 1730 (C=O). Polyurethane with different contents of DEM(OH)₂ (0 to 25 wt %) is designated as PU-Dx, where x is the weight percentage of DEM(OH)₂ determined by elemental analysis (Table S1). The characterization data are summarized in Table 1.

3. CHARACTERIZATION

3.1. Proton Nuclear Magnetic Resonance Spectroscopy (¹H NMR).

All ¹H NMR spectra were recorded on a Bruker

AV600 NMR spectrometer using CDCl₃ as the solvent and tetramethylsilane (TMS) as the internal standard.

3.2. Fourier Transform Infrared Spectroscopy (FTIR).

FTIR spectra were recorded on a Bruker VECTOR-22 IR spectrometer. The spectra were collected from 64 scans with a spectral resolution of 4 cm⁻¹ by the KBr disk method.

3.3. High-Resolution Mass Spectrometry (HRMS).

HRMS was performed by using electrospray ionization (ESI) and recorded on a Bruker maXis impact.

3.4. Differential Scanning Calorimetry (DSC).

The DSC measurement was performed on a NETZSCH DSC 200F3 differential scanning calorimeter under a nitrogen flow of 50 mL/min. Each sample was quickly heated to 110 °C and equilibrated at that temperature for 10 min, then it was then cooled to 25 °C and kept for 7 days before the measurement. All DSC traces were from the first heating process at a rate of 10 °C/min (Figure S1). The melting enthalpy change (Δ*H_m*) related to the crystallization of the PCL segment was assessed by the integration of the area of the melting peak. The crystallinity (*C_r*) was calculated from *C_r* = Δ*H_m*/(*W_{PCL}* × Δ*H_f*⁰) = Δ*H_m*/(50% × 136.08 J/g), where Δ*H_f*⁰ is the melting enthalpy change of PCL with fully crystallization.²⁶

3.5. Gel Permeation Chromatography (GPC).

The number-average molecular weight (*M_n*) and molecular weight distribution (*M_w*/*M_n*) were determined by GPC at 50 °C on a Waters 1515. A series of monodisperse polystyrenes were used as the standard in the fluent of DMF/LiBr solution (0.05 M in LiBr) at a flow rate of 1.0 mL/min.

3.6. Elemental Analysis. The atomic concentration (C, N, H, or S) of PU-Dx was measured with a Vario EL cube elemental analyzer (Elementar, Germany). The weight percentage of DEM was estimated from the sulfur content.

3.7. Contact Angle (CA) Measurements. CA measurements were conducted on an OCA 40 Micro (DataPhysics,

Germany) by depositing 3 μL liquid droplets on a PU-Dx surface using the sessile method. Contact angles of water and diiodomethane were measured at 25 $^{\circ}\text{C}$ to calculate the surface energy using the Owens–Wendt–Rabel–Kaelble method. Each CA value was the average of those from five repeated measurements.

3.8. Atomic Force Microscope (AFM) Measurements.

The PU-Dx film formed on a gold-coated quartz crystal surface was imaged on an XE-100 AFM (Park Systems, Korea) in air and ASW. The surface morphology was obtained in noncontact mode with an NCHR cantilever (spring constant 42 N/m). The rms roughness (R_q) was obtained on the basis of the images from three different locations (scan size 5.0 $\mu\text{m} \times 5.0 \mu\text{m}$, scan rate 1 Hz) using AFM analysis software.

3.9. Surface Zeta Potential Measurements. The surface zeta potential (ζ) was measured by using a SurPASS electrokinetic analyzer (Anton Paar, Austria). ζ at different pH values was determined with a SurPASS adjustable gap cell in the presence of ASW or fresh water by using pH-electrometric titration. The PU-Dx film was rinsed with electrolyte to equilibrate surface swelling before the test. ζ was calculated from streaming potential measurements according to

$$\zeta = \frac{dU}{dp} \times \frac{\eta}{\epsilon \times \epsilon_0} \times k_B$$

where dU/dp is the slope of streaming potential versus pressure, η is the electrolyte viscosity, ϵ_0 is the permittivity, ϵ is the dielectric coefficient of the electrolyte, and k_B is the electrolyte conductivity.

3.10. Quartz Crystal Microbalance with Dissipation (QCM-D). The adsorption of proteins and enzymatic degradation were monitored with QCM-D. The PU-Dx film was prepared by spin-casting PU-Dx solution in THF solution (5.0 mg/mL) on a spin-coater (CHEMAT, KW-4) at 4000 rpm in air. The thickness values of the films determined by spectroscopic ellipsometry are close to each other (~ 40 – 50 nm). QCM-D and the AT-cut quartz crystal with a fundamental resonance frequency of 5 MHz were from Q-sense AB (Sweden). The quartz crystal was mounted in a fluid cell with one side exposed to the solution. The effects of surface roughness were minimized by using highly polished crystals with a root-mean-square roughness of less than 3 nm. The details of the QCM-D measurements can be found elsewhere.²⁷ Briefly, the mass of a thin layer on a quartz crystal relates to the decrease in the resonance frequency of the crystal, whereas the dissipation factor is related to the viscoelastic properties of the additional layer. In the present study, ASW or fresh water was used as the reference. The protein solution (1.0 mg/mL) or lipase PS solution (0.5 mg/mL) was delivered to the surface at a flow rate of 150 $\mu\text{L}/\text{min}$. The changes in frequency (Δf) and dissipation (ΔD) gave information about the protein adsorption or degradation and the structural change of the PU-Dx films. All of the experiments were performed at 25 $^{\circ}\text{C}$, and the presented data were from the third overtone ($n = 3$).

3.11. Surface Plasmon Resonance (SPR). SPR measurements were performed on an SPR Navi 210A (Bionavis) instrument equipped with an autosampler accessory at 25 $^{\circ}\text{C}$. The amount of mass change on the gold sensor slide can be followed by monitoring the change in angular position (angular scan mode) over time.²⁸ The SPR angle shift is linearly related to the added mass of the layer with $0.1^{\circ} \approx 1 \text{ ng}/\text{mm}^2$. The

protein solution (1.0 mg/mL) was delivered to the surface at a flow rate of 10 $\mu\text{L}/\text{min}$.

3.12. Antibacterial Assays. *Micrococcus luteus* (*M. luteus*), a marine bacterium, was used to evaluate the antibacterial adhesion characteristics of the polyurethane coatings. A bacterial suspension was prepared with ASW at a concentration of 10^9 cells/mL before the measurement. Each substrate with the PU-Dx film was then immersed in 50 μL of the bacterial suspension under static conditions at 30 $^{\circ}\text{C}$ for 60 min. The surface of the PU-Dx film after immersion in the bacterial suspension was stained with a LIVE/DEAD BacLight bacterial viability kit. The adhered bacterial cells were examined with a fluorescence microscope (Scope A1, Zeiss).

4. RESULTS AND DISCUSSION

We first investigated the hydrophilicity of the polymer surface. Figure 1 shows the contact angle (CA) of water and the surface

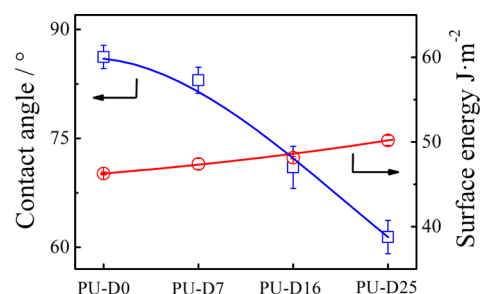


Figure 1. Contact angle (CA) of water and surface energy of the PU-Dx surface at 25 $^{\circ}\text{C}$.

energy of the PU-Dx surface. We can see that CA decreases as the DEM content increases, indicating that the incorporation of DEM increases the hydrophilicity. On the other hand, the surface energy slightly increases from ~ 34 (PU-D0) to ~ 31 (PU-D25), indicating that the surface becomes smoother as the DEM content increases. Once the PU-Dx film is immersed in ASW (Figure 2b), R_q increases in comparison to that in air due to the swelling and reorientation of the film induced by water.²⁹ Note that R_q for PU-D0 slightly varies because it does not contain hydrophilic DEM. Anyhow, PU-Dx has a relatively smooth surface in ASW.

Figure 2a shows the surface morphology and rms roughness (R_q) of PU-Dx observed by AFM in air. For PU-D0, R_q is ~ 0.34 , indicating a smooth surface. When DEM groups are incorporated, R_q changes from ~ 0.34 (PU-D0) to ~ 0.31 (PU-D25), indicating that the surface becomes smoother as the DEM content increases. Once the PU-Dx film is immersed in ASW (Figure 2b), R_q increases in comparison to that in air due to the swelling and reorientation of the film induced by water.²⁹ Note that R_q for PU-D0 slightly varies because it does not contain hydrophilic DEM. Anyhow, PU-Dx has a relatively smooth surface in ASW.

We studied protein adsorption on the PU-Dx surface by QCM-D and SPR to evaluate their protein resistance. Figure 3 shows the time dependence of the frequency shift (Δf) and the energy dissipation shift (ΔD) for the adsorption of BSA on PU-Dx surfaces in ASW. For PU-D0, after BSA is introduced, Δf sharply decreases and then gradually levels off. After rinsing with ASW, Δf still has a marked decrease in comparison to the baseline. It is known that the increase in mass on the sensor surface causes the frequency to decrease.³⁰ The decrease in Δf indicates the adsorption of BSA on the surfaces. On the other hand, ΔD increases with the thickness but decreases with the rigidity of a layer. Namely, a dense and rigid structure leads to a small dissipation of energy, whereas a looser and thicker structure results in a larger dissipation.³¹ The marked increase

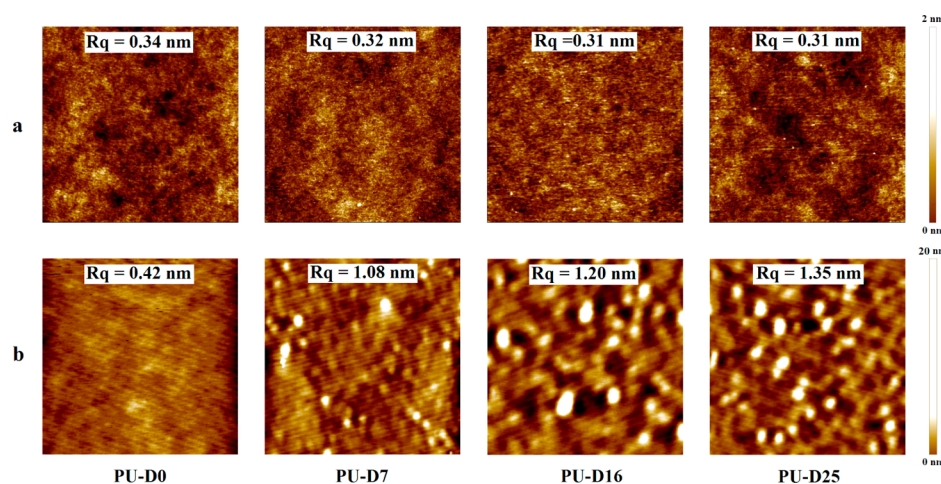


Figure 2. AFM images of PU-D x surfaces in air (a) and in artificial seawater (b). The scan size was $5\ \mu\text{m} \times 5\ \mu\text{m}$.

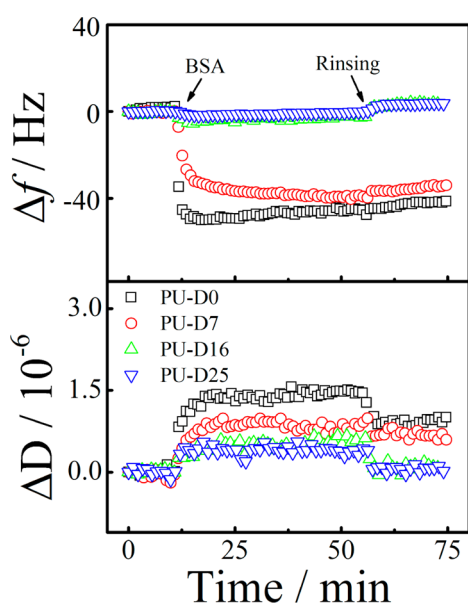


Figure 3. Time dependence of the QCM-D frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of BSA on the PU-D x surface in artificial seawater at $25\ ^\circ\text{C}$.

in ΔD relative to the baseline further indicates that BSA is adsorbed, forming a viscoelastic layer. When the DEM content is above 16 wt % (PU-D16 or PU-D25), Δf and ΔD almost return to their baselines after rinsing, indicating that no BSA is adsorbed. Clearly, the protein resistance of PU-D x to BSA increases with DEM content. Because the PU-D x surface becomes hydrophilic as the DEM content increases (Figure 1), its protein resistance should be associated with surface wettability. Actually, because of the lower DEM content, the PU-D0 or PU-D7 surface is more hydrophobic and has more protein adsorption. Note that the effects of surface roughness and thickness on protein resistance are not considered when we compare different samples because their rms roughness and thickness are similar (Figure 2). Previous studies also demonstrated that the surface topography has a slight effect on the protein resistance of polyurethane.^{32,33} Accordingly, the surface wettability plays a critical role in the protein resistance of PU-D x .

Figure 4 shows the time dependence of the SPR angle shift for the adsorption of BSA on the PU-D x surface in ASW. For

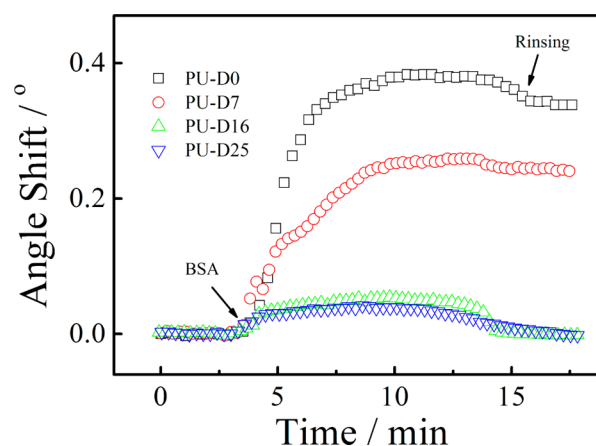


Figure 4. Time dependence of the SPR angle shift for the adsorption of BSA on the PU-D x surface in artificial seawater at $25\ ^\circ\text{C}$.

PU-D0 or PU-D7, a significant angle shift can be observed when BSA is introduced. After rinsing with ASW, some weakly attached proteins are removed, and the angle shifts are 0.33 and 0.24° relative to the baseline, respectively. When the DEM content is above 16 wt % (PU-D16 or PU-D25), the angle shift returns to the baseline after rinsing, indicating that almost no BSA is adsorbed. Clearly, the resistance of PU-D x to BSA adsorption increases with DEM content. This agrees with the QCM-D results. The adsorbed mass of BSA on the PU-D x surface can be estimated by the SPR shift. For PU-D0, PU-D7, PU-D16, and PU-D25 they are 3.3 , 2.4 , 0 , and $0\ \text{ng}/\text{mm}^2$, respectively. They can also be estimated from QCM-D measurements in terms of the Voigt model.³⁴ For PU-D0, PU-D7, PU-D16, and PU-D25, they are 3.5 , 2.6 , 0 , and $0\ \text{ng}/\text{mm}^2$, respectively. Clearly, the mass obtained by SPR is less than that from QCM-D. This is understandable because the latter is a hydrodynamic mass that includes those of protein and the coupled water,²⁷ whereas the former is the mass of the protein.

We have also examined the adsorption of fibrinogen and lysozyme on PU-D x surfaces by the use of QCM-D and SPR (Figures S2 and S3). Our studies demonstrate that the resistance of PU-D x to the adsorption of fibrinogen or

lysozyme increases with DEM content. When the DEM content is 25 wt %, no such a protein is adsorbed on the PU-D α surface.

To be clear about the protein-resistance mechanism, we measured the surface potential for PU-D25 at different pH and salinity values. Figure 5 shows the ζ potential of PU-D25 in

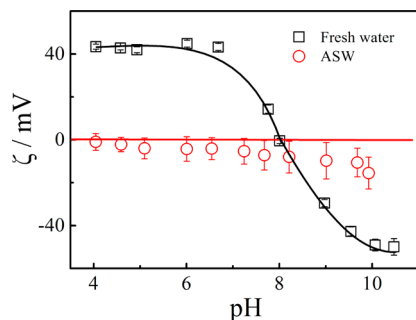


Figure 5. Surface ζ potential of PU-D25 in fresh water and ASW as a function of pH at 25 °C.

fresh water and ASW as a function of pH at 25 °C. In fresh water, the ζ potential varies with pH. The isoelectric point (IEP) of the PU-D25 surface is located at ~ 8.0 , close to the pK_a of PDEM reported before.³⁵ At pH < 8.0 , the PU-D α surface is positively charged, and the ζ -potential value decreases with pH, indicating the ongoing deprotonation of the tertiary amine in the DEM pendant group. At pH > 8.0 , the surface is negatively charged and the ζ -potential value increases with pH. Note that DEM is uncharged at pH > 8.0 ; the negative value is probably due to the preferential absorption of negative charges such as OH^- on the surface although this explanation remains controversial.³⁶ In ASW with pH ~ 8.2 , the ζ potential of the PU-D25 surface is close to zero. This is understandable because DEM groups are fully deprotonated. On the other hand, the ζ potential of the PU-D25 surface is also close to zero at pH < 8.0 , indicating that the charges are completely screened due to the high salinity in ASW.

To have a better understanding of the protein-resistance mechanism, we also investigated the effects of pH and salinity on protein adsorption. Figure 6 shows the changes in the QCM-D frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen, BSA, or lysozyme on PU-D25 surface in fresh water and ASW as a function of pH. For fibrinogen, both Δf and ΔD have the largest shift at pH 6.0, and a

significant decrease can be observed in fresh water at pH 4.0, 8.2, or 10.0, indicating that the protein adsorption of the PU-D25 surface has a pH dependence. Consider that the IEPs of PU-D25 and fibrinogen are 8.0 and 5.5, respectively. At pH 4.0, both of them are positively charged, so the electrostatic repulsion leads to protein resistance. At pH 6.0, PU-D25 and fibrinogen are oppositely charged, and the electrostatic attraction results in the largest protein adsorption. At pH 8.2 or 10.0, PU-D25 is uncharged, so the protein adsorption decreases significantly. For fibrinogen in ASW, both Δf and ΔD are close to zero after rinsing regardless of pH, indicating that PU-D25 surfaces can effectively inhibit the adsorption of the protein in ASW. This is because the tertiary amines in DEM groups at pH > 8.0 are fully deprotonated, the charges are screened in high-salt solution at pH < 8.0 , and there is no electrostatic attraction between fibrinogen and PU-D25. The adsorption of BSA exhibits a similar trend because its IEP is 4.8. No adsorption of lysozyme on the PU-D25 surface happens in either fresh water or ASW in the range of pH we investigated. This is because lysozyme with IEP ~ 11.0 and the PU-D25 surface are positively charged in the range of pH 4–8, and electrostatic repulsion leads to protein resistance. At pH 8–10, the PU-D25 surface does carry net charges but is hydrophilic, and the hydrophilicity of the polymer and the net neutral charge give rise to the protein resistance.

Generally, the protein adsorption of a material is mainly determined by the surface wettability, surface energy, surface topography, and surface chemistry.^{37,38} In the present study, the samples have similar surface energies and surface topographies, so the surface wettability and surface chemistry play critical roles in the protein resistance. In other words, the hydrophilicity and charging of the substrate surface profoundly influence the ability to resist protein adsorption. As discussed before, a protein-resistant surface should have the following characteristics: (1) hydrophilicity, (2) the ability to accept hydrogen bonds, (3) the inability to donate hydrogen bonds, and (4) a net neutral charge.³⁷ In this study, the pendant DEM groups are hydrophilic. As the DEM content increases, the hydrophilicity of the polymer surface increases so that the protein resistance of the polymer increases. On the other hand, the charging of DEM depends on the pH and salinity.^{39–41} In fresh water, the tertiary amine in DEM is protonated at pH < 8.0 so that the polymer can adsorb proteins with negative charges. In seawater, the tertiary amines in DEM groups are fully deprotonated at pH > 8.0 , and the charges are screened in

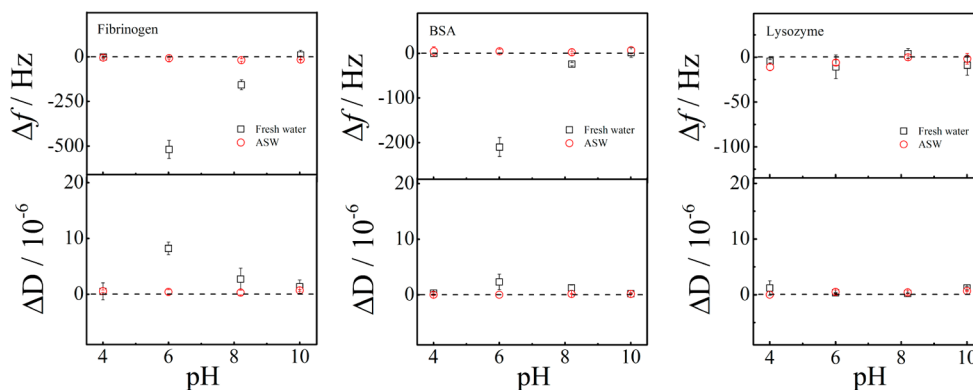


Figure 6. Changes in the QCM-D frequency shift (Δf) and energy dissipation shift (ΔD) for the adsorption of fibrinogen, BSA, or lysozyme on the PU-D25 surface in fresh water and artificial seawater at 25 °C as a function of pH.

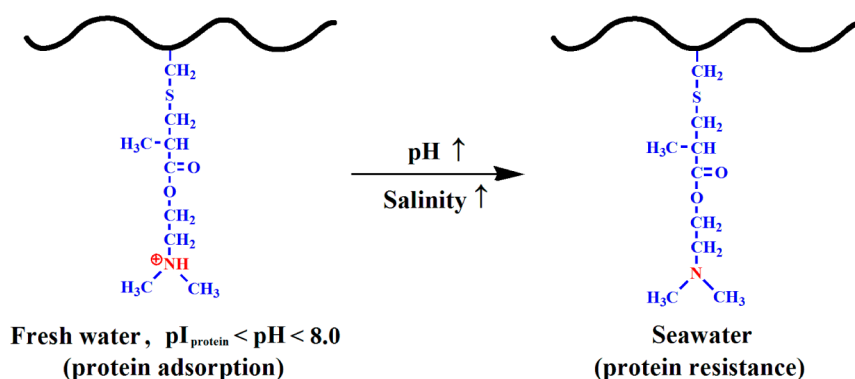


Figure 7. Protein-resistance mechanism of PU-Dx in seawater.

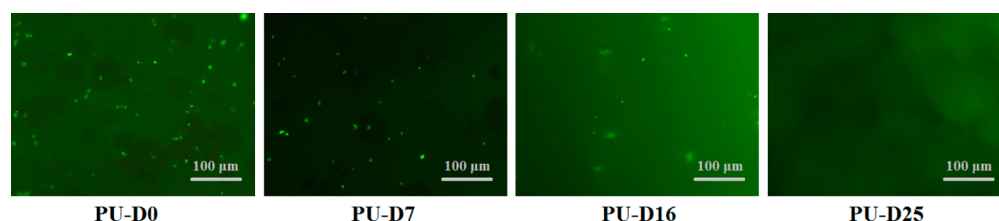


Figure 8. Fluorescence microscopy images of the PU-Dx surfaces after exposure to *M. luteus* (10^9 cells/mL) in artificial seawater at 30 °C for 60 min.

high-salt solution at $\text{pH} < 8.0$. Thus, PU-D25 is hydrophilic and does not carry net charges in seawater, so it could form the hydration layer to resist protein adsorption, similar to the behavior of zwitterionic polymers.⁴² The mechanism is illustrated in Figure 7.

We also investigated the antibacterial ability of PU-Dx. Figure 8 shows fluorescence microscopy images of the PU-Dx surfaces after exposure to *M. luteus* (10^9 cells/mL) in ASW for 60 min. For PU-D0, a large number of living bacterial cells readily adhered to the surface; namely, the surface is highly susceptible to bacterial adhesion and colonization. However, as the DEM content increases, an obvious reduction in bacterial adhesion was observed (PU-D7 and PU-D16). As the DEM content increases up to 25 wt % (PU-25), almost no bacterial cells can be observed, indicating that the surface has good antibacterial adhesion ability. This is because the bacterial adhesion generally begins with the adsorption of extracellular polymeric substances (EPS) mainly consisting of proteins and glycoproteins. A protein-resistant surface can reduce the initial attachment and delay the colonization of bacteria.^{43–45} Note that quaternized DEM can effectively kill bacteria by disrupting the cytoplasmic membrane because it is positively charged and the bacterial surfaces are negatively charged.^{46,47} In seawater at $\text{pH} \sim 8.2$ and high salinity, DEM does not carry net charges, so the antibacterial ability comes from its protein resistance. On the other hand, considering that a large population of microorganisms such as bacteria, actinomycetes, and fungi in natural marine environments can produce enzymes,^{48,49} we also examined the degradation of the polymer in the presence of lipase PS to evaluate the enzymatic degradation in a marine environment. Our results indicate the PU-Dx is able to degrade in a marine environment and that the degradation rate increases with DEM content (Figure S4).

5. CONCLUSIONS

We have prepared polyurethane with poly(ϵ -caprolactone) (PCL) segments in the main chain and 2-(dimethylamino)

ethyl methacrylate (DEM) as pendant groups. The polymer exhibits protein resistance in seawater but not in fresh water. The adsorption of proteins in fresh water is due to the electrostatic attractions between polymer and the proteins. The hydrophilicity of the polymer and the net neutral charge in seawater are responsible for the protein resistance. The polymer also exhibits antibacterial ability and biodegradation in a marine environment. It may find applications in marine antibiofouling.

■ ASSOCIATED CONTENT

● Supporting Information

Elemental analysis, DSC curves, and protein adsorption and degradation of PU-Dx. The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.langmuir.5b01720.

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Notes

The authors declare no competing financial interest.

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